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Communication

Iron-Catalyzed Contrasteric Functionalization of Allenic C(sp^2)–H Bonds: Synthesis of α-Aminoalkyl 1,1-Disubstituted Allenes

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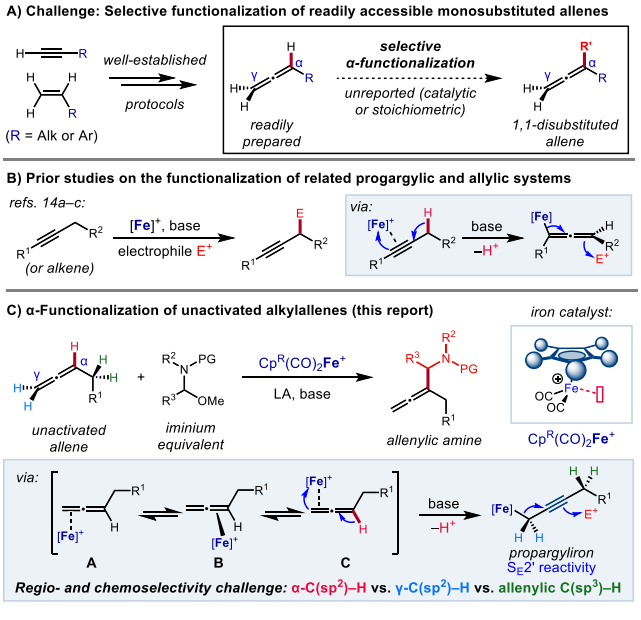
ABSTRACT: An iron-catalyzed C–H functionalization of simple monosubstituted allenes is reported. An efficient protocol for this process was made possible by the use of a newly developed electron-rich and sterically hindered cationic cyclopentadienyliron dicarbonyl complex as the catalyst and *N*-sulfonyl hemiaminal ether reagents as precursors to iminium ion electrophiles. Under optimized conditions, the use of a mild, functional-group-tolerant base enabled the conversion of a range of monoalkyl allenes to their allenylic sulfonamido 1,1-disubstituted derivatives, a previously unreported and contrasteric regiochemical outcome for the C–H functionalization of electronically unbiased and directing-group-free allenes.

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As a geometrically unique and synthetically versatile substructure,[1] the allene occupies a prominent position in chemical synthesis. Allene derivatives serve as important starting materials for catalytic transformations, including ring forming processes,[2] hydrofunctionalization and difunctionalization reactions,[3] and various rearrangements.[4] In addition, they occur in a diverse range of bioactive natural products and pharmaceuticals.[5] An array of general synthetic methods are available to efficiently prepare substituted allenes,[6] including a number of recently developed catalytic protocols.[7] Moreover, monosubstituted derivatives are particularly accessible, as they can be prepared by the homologation of widely available alkenes and alkynes.[8] The C–H functionalization of monosubstituted allenes is thus an attractive approach for the preparation of more complex, functionalized derivatives. In this regard, allenes bearing π-electron-withdrawing groups have been elaborated at the α-C(sp^2) position using (aza-)Morita–Baylis–Hillman-type or aldol/Mannich-type reactions.[9]

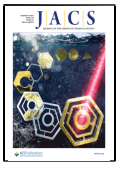
The C–H functionalization of simple alkyl or aryl allenes, however, represents a daunting synthetic challenge, especially in a catalytic sense (Scheme 1A). While formal C–H functionalization reactions of allenes that deliver various olefinic products are well established,[10] only a handful of protocols allow for the C–H functionalization of electron-neutral allenes without accompanying addition to or isomerization of the sensitive allene moiety. The reported processes that retain the allene moiety require the use of polysubstituted or specialized allenes as starting materials.[11] Although the direct lithiation of monosubstituted allenes followed by electrophilic quench is possible, functional group tolerance is limited, and, in the absence of an α-heteroatom directing group,[12] these procedures generally give the 1,3-difunctionalized product with moderate to high levels of regioselectivity.[13] To the best of our knowledge, a synthetic procedure, catalytic or stoichiometric, for the C–H functionalization of electronically neutral monosubstituted allenes to selectively afford 1,1-disubstituted products has not yet been disclosed. Such a process would be highly desirable, given the limited number of synthetic procedures that give this substitution pattern.

Scheme 1. Access to Allenylic Amines by C–H Functionalization



Previously, we reported a strategy for catalytic propargylic and allylic C–H functionalization wherein a cationic cyclopentadienyliron dicarbonyl complex was employed to enhance

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the acidity of α-C−H bonds adjacent to the iron-complexed C−C π-bond (Scheme 1B).[14] The acidity enhancement is such that allylic and propargylic C−H bonds ($pK_a \approx 35$ to 45) become acidic enough to be removed by weak pyridine or alkylamine bases ($pK_{aH}^{aq.} \leq 12$). The η^1-allyl- and allenyliron species generated therefrom react with a range of electrophilic species with S_E2' selectivity to give the α-C−H functionalized product.

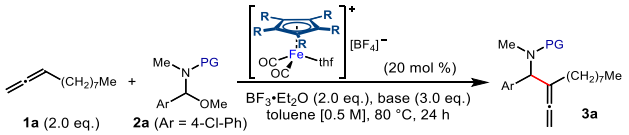
We anticipated that this platform could be applied to the functionalization of allenes through the analogous enhancement of the acidity of the allenic C−H bond (Scheme 1C). Studies by the groups of Rosenblum and Pettit suggest that regio- and stereoisomeric cationic iron−allene complexes equilibrate and could potentially lead to the formation of a mixture of isomeric functionalization products.[15] We hypothesized that, under catalytic conditions, deprotonation and subsequent S_E2' reaction of the stereoelectronically well-disposed isomer **C** would be selective,[15b] ultimately leading to formation of the *contrasteric* C−H functionalization product.

Given the presence of the allenylic amine substructure in a variety of pharmacologically active scaffolds,[16] we sought to apply this approach to a synthesis of this class of compounds through the interception of the proposed propargyliron intermediate with *in situ* generated iminium electrophiles. In this Communication, we report the successful development of the proposed C−H functionalization process, which ultimately led to new catalysts and reagents for a scalable and operationally simple catalytic protocol for preparing 1,1-disubstituted allenes bearing a sulfonamide function at the allenylic position.

We commenced our development of an allene functionalization process by exploring a range of *N*,*O*-acetal reagents, in the presence of $BF_3{\cdot}Et_2O$ as a Lewis acid, for the generation of iminium electrophiles.[17] Our initial investigations, using the previously reported combination of 2,6-lutidine as the base and $[Cp^*Fe(CO)_2(thf)]^+BF_4^-$ as the catalyst, revealed that *N*-carbonylmethoxy and *N*-benzoyl protected *N*,*O*-acetals were ineffective coupling partners (Table 1, entries 1−2). However, the use of the *N*-tosyl derivative afforded a moderate yield of the allenylic sulfonamide (entry 3). The yield of the desired product could be improved by tuning the arylsulfonyl group of the reagent, with better results obtained for more electron-donating aryl groups (entries 3−6). Next, the effect of the base was investigated. Among 2,6-dimethylpyridine derivatives, a clear trend emerged in which weaker bases led to higher observed yields. In particular, 4-bromolutidine ($pK_{aH}^{aq.} \approx 5.3$)[18] was found to be a particularly effective base among a variety of heterocyclic amines and alkylamines evaluated (entries 6−11).

After further unsuccessful efforts to optimize the reaction by variation of base, *N*-protecting group, Lewis acid, and other reaction parameters, we turned to the development of new catalysts for further improvements in reactivity. We posited that more rapid equilibration of cationic iron−allene species would be beneficial (Scheme 1C). On the basis of our previous finding that analogous cationic iron−alkyne complexes dissociate more rapidly when bulky, electron-rich cyclopentadienyl ligands are used,[14b] we prepared novel catalysts bearing Et_5C_5 and $^nPr_5C_5$ ligands.[19] As we had hoped, the use of these bulkier ligands provided additional increases in the yield (entries 12−14). A final adjustment of base stoichiometry allowed the allenylic sulfonamide to be isolated in high yield (79%, entry 15).

Table 1. Optimization of Allenylic Amine Synthesis



Entry	R	Base	PG	Yield (%)[a]
1	Me	2,6-lutidine	CO_2Me	0
2	Me	2,6-lutidine	Bz	0
3	Me	2,6-lutidine	Ts	27
4	Me	2,6-lutidine	Ns	21
5	Me	2,6-lutidine	Mbs	32
6	Me	2,6-lutidine	Mts	39
7	Me	TMPH	Mts	5
8	Me	2,4,6-collidine	Mts	21
9	Me	4-Cl-2,6-lutidine	Mts	50
10	Me	4-Br-2,6-lutidine	Mts	55
11	Me	4-Br-2-picoline	Mts	0
12[b]	Me	4-Br-2,6-lutidine	Mts	63
13[b]	Et	4-Br-2,6-lutidine	Mts	73
14[b]	^{n}Pr	4-Br-2,6-lutidine	Mts	77
15[bc]	^{n}Pr	4-Br-2,6-lutidine	Mts	82 (79)[d]

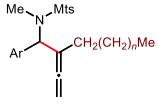
[a]Yields were determined by ^{1}H NMR using 1,1,2,2-tetrachloroethane as the internal standard. [b]$PhCF_3$ as solvent and at 100 °C. [c]4.0 equiv base. [d]Isolated yield (0.2 mmol scale). TMPH = 2,2,6,6-tetramethylpiperidine, Mts = 2,4,6-trimethylbenzenesulfonyl, Mbs = 4-methoxybenzenesulfonyl.

With these reaction conditions in hand, we investigated the scope and applicability of the allene functionalization reaction (Table 2, top). Pleasingly, a wide range of monosubstituted alkyl allenes bearing an array of functional groups was found to afford the desired allenylic sulfonamides in moderate to good yields (30 examples, 34−80% yield). In particular, allenes bearing an alkyl chloride (**3f**), an alkyl tosylate (**3h**), a boronic acid pinacol ester (**3k**), a nitro group (**3l**), a thioether (**3m**), and a phthalimido group (**3u**) were all successful substrates, as were several protected allene alcohols, including an acetate (**3o**), a benzoate (**3p**), and a *tert*-butyldimethylsilyl ether (**3t**). Allenes with secondary (**3d**) or β-branched (**3c**) alkyl substituents also reacted efficiently. A number of substrates containing heterocycles or pharmaceutical fragments (**3q**−**3s**, **3y**−**3ae**) were also suitable starting materials. As a result of the mild base used in this transformation, enolizable esters (**3g** and **3o**) likewise gave good results under the optimized conditions. We then examined substrates **1v** and **1w** to determine whether the allene group would undergo selective functionalization in the presence of a terminal alkene or aryl alkyne. In the event, both starting materials underwent the desired allenic functionalization reaction with complete selectivity (>20:1) over functionalization of the alkene or alkyne, as determined by ^{1}H NMR analysis of the crude material. These results are consistent with the observation that 4-bromolutidine was unsuccessful for allylic and propargylic functionalization and that stronger bases are needed for functionalization at these positions.[14a,b] Finally, we note that an aryl allene (**3x**) provided the desired product in diminished but still useful yield.

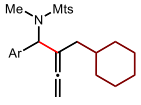
https://doi.org/10.1021/jacs.1c07512
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Table 2. Scope of the Allenylic Amine Synthesis[a]

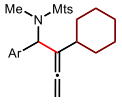
The scope of allenes (Ar = 4-ClC$_6$H$_4$)



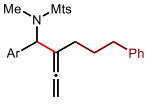
3a, *n* = 6, 79% yield
(71% yield, 3.0 mmol scale [*b*])
3b, *n* = 8, 76% yield



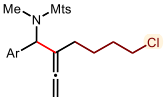
3c, 80% yield



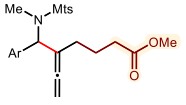
3d, 70% yield



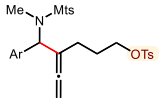
3e, 65% yield



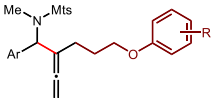
3f, 71% yield



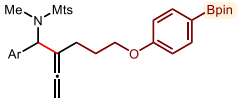
3g, 69% yield



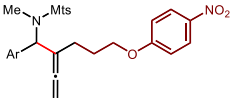
3h, 64% yield



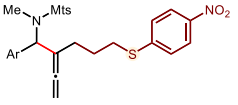
3i, R = 4-Br, 75% yield
3j, R = 2,6-Me$_2$, 67% yield



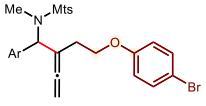
3k, 62% yield



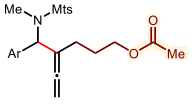
3l, 70% yield



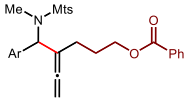
3m, 40% yield [*c*]



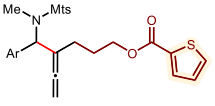
3n, 67% yield



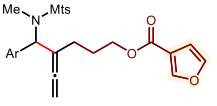
3o, 73% yield



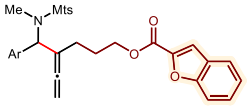
3p, 65% yield



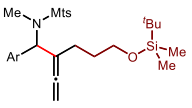
3q, 71% yield



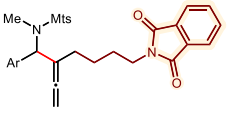
3r, 68% yield



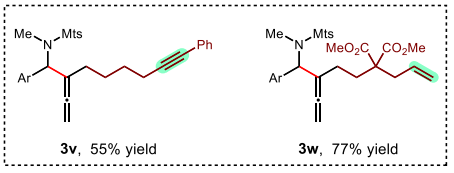
3s, 73% yield



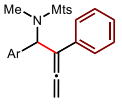
3t, 61% yield



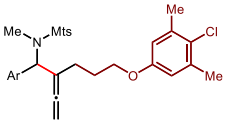
3u, 73% yield



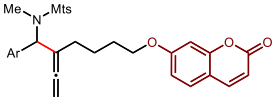
3v, 55% yield
3w, 77% yield



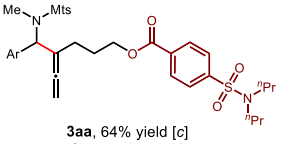
3x, 34% yield



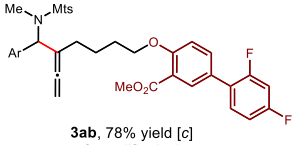
3y, 69% yield
from chloroxylenol



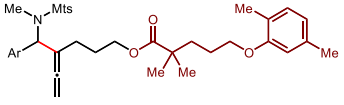
3z, 62% yield [*c*]
from umbelliferone



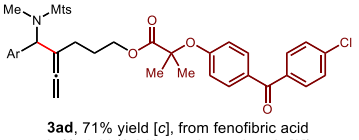
3aa, 64% yield [*c*]
from probenecid



3ab, 78% yield [*c*]
from diflunisal

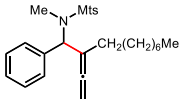


3ac, 66% yield [*c*]
from gemfibrozil

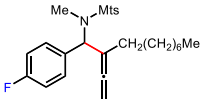


3ad, 71% yield [*c*], from fenofibric acid
65% yield (2.348 g, *5.0 mmol scale* [*b*])

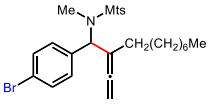
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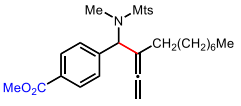
4a, 65% yield



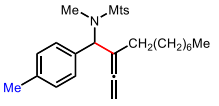
4b, 70% yield



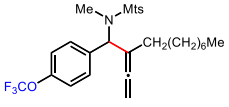
4c, 75% yield



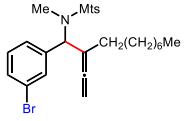
4d, 77% yield



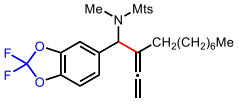
4e, 50% yield



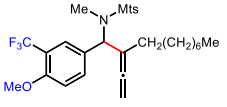
4f, 75% yield



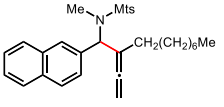
4g, 69% yield



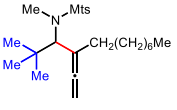
4h, 71% yield



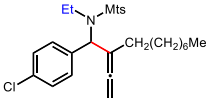
4i, 52% yield



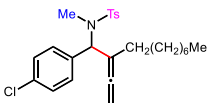
4j, 61% yield



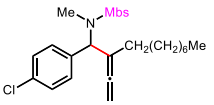
4k, 88% yield
85% yield (1.5 mmol scale [*b*])



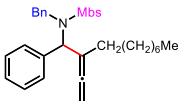
4l, 69% yield



4m, 58% yield



4n, 66% yield



4o, 59% yield

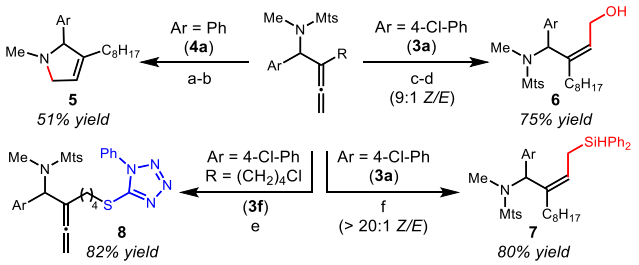
[a]Conditions: **1** (2.0 equiv), **2** (0.2 mmol), [(^{n}Pr$_5$C$_5$)Fe(CO)$_2$(thf)]$^+$BF$_4^-$ (20 mol %), BF$_3$·Et$_2$O (2.0 equiv), 4-bromolutidine (4.0 equiv), PhCF$_3$ (0.5 M), 100 °C. [b]15 mol % catalyst. [c]**1** (0.2 mmol), **2** (1.5 equiv). Mts = 2,4,6-trimethylbenzenesulfonyl, Mbs = 4-methoxybenzenesulfonyl.

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Subsequent exploration of the scope of the *N*,*O*-acetal (Table 2, bottom) revealed that coupling partners carrying electron-neutral (e.g., **4a**, **4b**) to moderately electron-poor (e.g., **4d**, **4f**) aryl groups furnished good or very good yields of the desired product. Replacement of the aryl group with a *tert*-butyl group (**4k**) also resulted in a successful outcome. However, lower yields were observed when electron-rich aryl groups were used (e.g., **4e**). Finally, the transformation retained its efficiency when the methyl group on the nitrogen was replaced with larger alkyl groups (**4l**, **4m**, **4o**) or when the protecting group was switched to the 4-methoxybenzenesulfonyl group (**4n**). In all cases examined, the 1,1-disubstituted allenylic sulfonamide was formed exclusively over other possible regioisomers (>20:1 by ^{1}H NMR analysis of the crude material).[20]

The protocol reported here could be performed on up to 5 mmol scale while maintaining synthetic efficiency (**3a**, **3ad**, and **4k**). Moreover, the products obtained were amenable to valuable downstream synthetic manipulations, including the removal of the sulfonyl protecting group in the presence of the potentially sensitive allene moiety (Scheme 2). In particular,

Scheme 2. Divergent Transformations of Products[a]

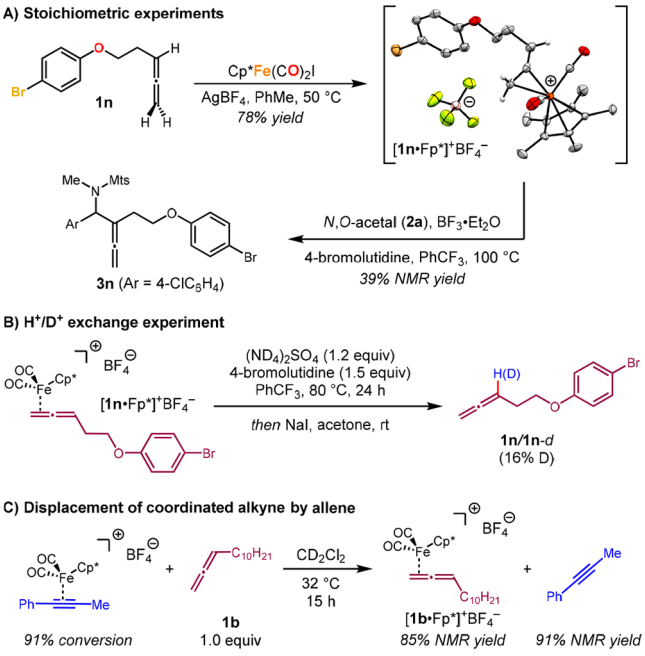


[a]Conditions: (a) Na/$C_{10}H_8$, THF, −78 °C, 8 h; (b) $Cu(OTf)_2$, CH_2Cl_2, rt, 24 h. (c) 9-BBN, THF, rt, 5 h; (d) H_2O_2, NaOH (3 M), 0 °C, 2 h. (e) 1-Phenyl-1*H*-tetrazole-5-thiol, K_2CO_3, KI, DMF, 90 °C, 24 h. (f) $Mo(CO)_6$, Ph_2SiH_2, PhMe, 110 °C, 36 h.

reductive removal of the sulfonyl group, followed by copper-catalyzed cyclization, afforded 1,2,3-trisubstituted 3-pyrroline derivative **5**.[21a,b] On the other hand, a regio- and stereoselective hydroboration−oxidation sequence afforded functionalized allylic alcohol (*Z*)-**6** in 75% yield (9:1 *Z*/*E*, crude),[21c] while Mo-catalyzed hydrosilylation afforded allylsilane (*Z*)-**7** in 80% yield (>20:1 *Z*/*E*, crude).[21d] Finally, nucleophilic substitution of alkyl chloride containing allenylic sulfonamide **3f** efficiently afforded a tetrazole-containing product (**8**).

We performed some preliminary investigations into the mechanistic aspects of this transformation. First, we prepared a cationic Fp*−allene complex by combining **1n** and Fp*I in the presence of $AgBF_4$ (Scheme 3A). In contrast to previous reports of the complexation of Fp^+ with 1,2-butadiene resulting in a 2:1 mixture of *syn* and *anti* stereoisomers, ^{1}H NMR data (rt to −55 °C) for [**1n**·Fp*]$^+BF_4^-$ indicated formation of a single major regio- and stereoisomer (>20:1 r.r. and d.r.). X-ray crystallographic analysis confirmed the expected *anti* stereochemistry (i.e., isomer **A** in Scheme 1C), along with a strongly bent allene geometry (149.7°), comparable to those of other allene−iron carbonyl complexes.[22] Exposure of [**1n**·Fp*]$^+BF_4^-$ to catalytically relevant conditions afforded **3n** in 39% yield. While allenyliron complexes were readily isolated upon treatment of Fp*−alkyne complexes with Et_3N,[14b] attempts to isolate a propargyliron complex by deprotonation of [**1n**·

Scheme 3. Selected Mechanistic Experiments



Fp*]$^+BF_4^-$ resulted in the slow formation of a complex mixture of iron-containing species over several hours.[23] To provide additional support for the proposed mechanism, [**1n**·Fp*]$^+BF_4^-$ was exposed to 4-bromolutidine in the presence of $(ND_4)_2SO_4$ as a D^+ source. Upon decomplexation, allene **1n** was found to have undergone H^+/D^+ exchange (16% D incorporation) exclusively at the allenic α-C(sp^2) position (Scheme 3b).[24]

In addition, the stabilities of the allene and alkyne complexes were also compared. Subjecting [η^2-(PhC≡CMe)Fp*]$^+BF_4^-$ to **1b** (1.0 equiv) at 32 °C in CD_2Cl_2 for 15 h resulted in the formation of [**1b**·Fp*]$^+BF_4^-$ in high conversion and yield (Scheme 3C). Moreover, when [Fp*(thf)]$^+BF_4^-$ was subjected to a mixture of PhC≡CMe and allene **1b** (2.0 equiv each) in CD_2Cl_2 (23 °C) in a competition experiment, formation of the alkyne complex was found to be kinetically favored (~1:2.5 ratio of allene/alkyne complexes at 10 min, <5% conversion), while the allene complex became predominant upon prolonged equilibration (>20:1 ratio of allene/alkyne complexes after 68 h, full conversion). Taken together, these results imply a larger kinetic barrier for the dissociation of an allene from the iron center compared to an alkyne, a factor of potential relevance to the superior performance of larger cyclopentadienyl ligands in the present catalytic transformation.

In summary, we have developed a novel deprotonative strategy for the C−H functionalization of terminal allenes, the implementation of which was enabled by a newly synthesized cationic cyclopentadienyliron dicarbonyl catalyst and the use of *N*-arylsulfonyl hemiaminal ether reagents as iminium electrophile precursors. Uniquely, this approach allowed for the heretofore unreported functionalization of the allene at the internal, sterically less-accessible C(sp^2)−H bond for the synthesis of 1,1-disubstituted allenes. Further mechanistic investigation and synthetic exploitation of the distinctive regio- and chemoselectivity of this system are ongoing in our laboratory.

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■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.1c07512.

Experimental procedures, spectroscopic data for the substrates and products, and crystallographic data for **3c** (CCDC 2091133) and [**1n**·Fp*]$^+$BF$_4^-$ (CCDC 2093855) (PDF)

Accession Codes

CCDC 2091133 and 2093855 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(20) A cyclic allene, cyclotrideca-1,2-diene, afforded coupling product with **2a** in moderate yield (35% NMR yield, 1:1 dr) when lutidine was used as the base, while an acyclic allene, 1,7-diphenylhepta-3,4-diene, was unreactive regardless of the base used (<5% NMR yield). In addition, *N,O*-acetals derived from enolizable aliphatic aldehydes (e.g., cyclohexanecarbaldehyde) do not give the desired allenylic sulfonamide product under the current conditions.

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